A research group investigated the glucogenic metabolism of threonine in rats.

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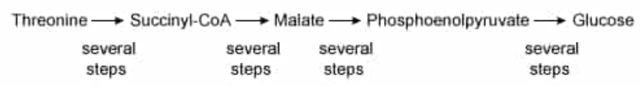


Scheme I

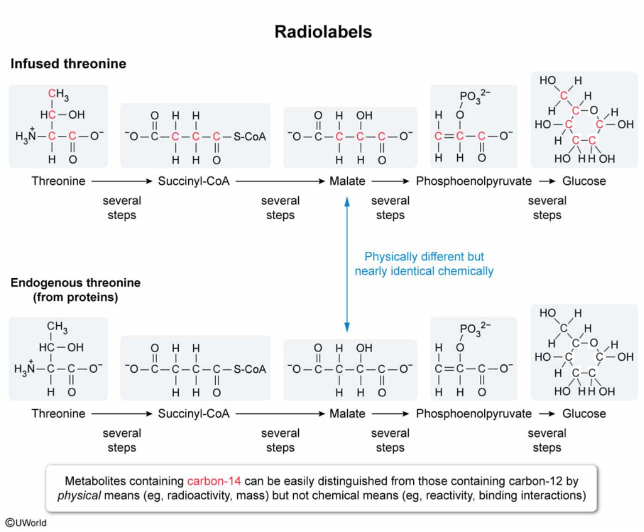
All carbons in the infused threonine were labeled with the radioactive isotope carbon-14. Levels of metabolic intermediates containing carbon-14 were monitored over time. The rate of incorporation of threonine carbons into glucose depended on the amount of time during which food was withheld prior to the threonine infusion. Rats that were deprived of food for 3 days produced labeled glucose at approximately twice the rate of rats that were deprived of food for 1 day.

Which of the following reasons does NOT describe why threonine was labeled with carbon-14 atoms?

- A. To distinguish between glucose molecules produced from infused threonine and those produced from endogenous sources. [4%]
- B. To improve chemical separation of metabolites derived from the infused threonine and metabolites derived from other sources. [69%]
- C. To facilitate detection of metabolic intermediates produced from the infused threonine. [5%]
- D. To confirm that the infused threonine molecules follow the proposed metabolic pathway. [20%]

Correct
68% answered correctly

Explanation:



Radioactive atoms are **commonly incorporated into molecules** used in studies of **metabolic pathways**. As the labeled molecules undergo metabolic reactions, the radioactive atoms are transferred from one pool of metabolites to another. Determining **which metabolites contain the radiolabel** at different time points allows researchers to **determine the steps of the pathway** used to form a given product and to **quantify reaction rates** and **concentrations**.

The question asks which reason does NOT describe why labeled threonine was used. Chemical separation of molecules requires significant differences in chemical properties that can be exploited (eg, different interactions with mobile or stationary phases in chromatography columns). Carbon-14 essentially has the same chemical properties as any isotope of carbon; therefore, its incorporation into threonine does *not* alter the overall chemical properties of threonine or its downstream metabolites. Consequently, incorporation of carbon-14 into these metabolites **does *not* allow them to be separated from unlabeled metabolites by chemical means**.

(Choice A) It is likely that rats will use fuel sources other than infused threonine (eg, endogenous proteins) as sources of glucose production. However, the endogenous proteins will *not* contain radioactive isotopes in significant quantities. Therefore, incorporation of carbon-14 *does* allow researchers to distinguish which glucose molecules were derived from the threonine they provided and which were not.

(Choices C and D) Metabolites can be isolated at different time points, and the extent of carbon-14 incorporation into these metabolites can be measured, allowing detection of intermediates that were derived specifically from the infused threonine. Molecules that incorporate carbon-14 at earlier time points must be produced prior to molecules that incorporate carbon-14 later, allowing characterization of the metabolic pathway that the molecules follow.

Educational objective:

Radioactive isotopes of atoms are commonly incorporated into molecules to track their progress through metabolic pathways and to monitor specific compounds. Different isotopes of an atom have essentially the same chemical properties.

Time spent: 0 sec	**QID**:403416
Subject: Biochemistry	**System:** Metabolic Reactions

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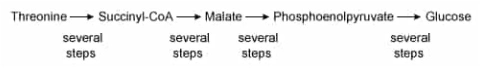


Scheme I

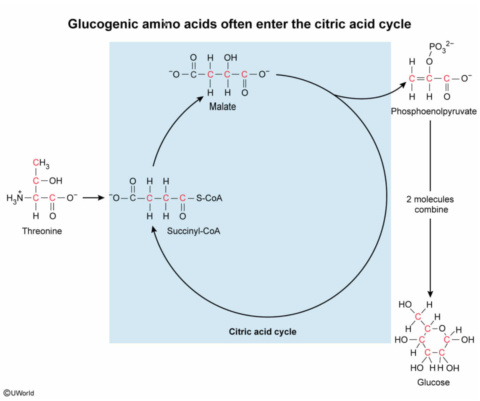
All carbons in the infused threonine were labeled with the radioactive isotope carbon-14. Levels of metabolic intermediates containing carbon-14 were monitored over time. The rate of incorporation of threonine carbons into glucose depended on the amount of time during which food was withheld prior to the threonine infusion. Rats that were deprived of food for 3 days produced labeled glucose at approximately twice the rate of rats that were deprived of food for 1 day.

Before the carbon atoms in threonine can be incorporated into glucose, they must enter

- A. glycogenesis [8%]
- B. the urea cycle [6%]
- C. glycogenolysis [6%]
- D. the citric acid cycle [79%]

Correct
78% answered correctly

Explanation:



The citric acid cycle is a metabolic process that combines oxaloacetate with acetyl-CoA to produce citrate. The citrate then undergoes several reactions that eventually convert it back to oxaloacetate. A new acetyl-CoA molecule can then be added, and the cycle repeats. Although this cycle is commonly depicted with acetyl-CoA as the entering molecule, many molecules can enter the cycle at other points (eg, as α-ketoglutarate or succinyl-CoA).

Scheme I shows that threonine can be converted to succinyl-CoA, which subsequently undergoes the steps of the cycle that convert it to malate. The malate is then able to undergo a series of reactions that eventually produce glucose. Because the reactions that convert succinyl-CoA to malate are part of the citric acid cycle, the carbon atoms in **threonine must enter the citric acid cycle** before they can be **converted to glucose**.

(Choice A) Glycogenesis is a series of reactions that convert glucose into glycogen. The carbon atoms in threonine can only enter glycogenesis *after* being incorporated into glucose, not before.

(Choice B) The urea cycle incorporates the *nitrogen* atoms from amino acids into urea for excretion. The carbon atoms from amino acids such as threonine are generally not involved in the urea cycle, and this process is not involved in glucose production.

(Choice C) Glycogenolysis is the degradation of glycogen to produce glucose. The carbon atoms in threonine could only enter glycogenolysis *after* being incorporated into glucose and then into glycogen.

Educational objective:
Metabolic pathways can be identified by the molecules they use as substrates.

Time spent:0 sec
Subject:
Biochemistry

QID:403417
System:
Metabolic Reactions

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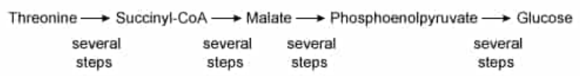

Scheme I

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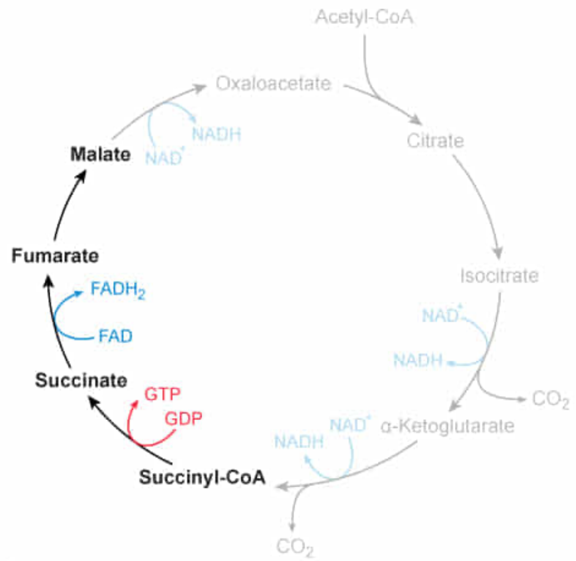
Which of the following describes the high-energy cofactors produced as the threonine carbons are converted from succinyl-CoA to malate?

- A. One GTP molecule is produced only. [5%]
- B. One GTP molecule and one NADH molecule are produced. [15%]
- ✔ C. One GTP molecule and one $FADH_2$ molecule are produced. [51%]
- D. One GTP molecule, one NADH molecule, and one $FADH_2$ molecule are produced. [27%]

Correct
51% answered correctly

Explanation:

High-energy intermediates of the citric acid cycle



The **citric acid cycle** includes several steps that produce **high-energy molecules** such as GTP, NADH, and $FADH_2$. GTP can then either be used directly in metabolic processes or it can be converted to ATP by transferring its γ-phosphate to ADP. NADH and $FADH_2$ enter the electron transport chain, which ultimately leads to more ATP production.

The question asks for the high-energy molecules produced as succinyl-CoA is converted to malate. Two steps within this process produce such molecules: conversion of succinyl-CoA to succinate produces GTP, and conversion of succinate to fumarate produces $FADH_2$. The conversion of fumarate to malate does *not* produce any high-energy molecules. Therefore, the given pathway produces **one GTP molecule and one $FADH_2$ molecule**.

(Choice A) This answer choice omits the $FADH_2$ that is produced when succinate is converted to fumarate.

(Choices B and D) NADH is produced during three steps of the citric acid cycle, but none of those steps occur during the conversion of succinyl-CoA to malate.

Educational objective:
The citric acid cycle produces three NADH molecules, one GTP molecule, and one $FADH_2$ molecule per round. Each molecule is produced in a distinct step of the cycle, and a given subset of cycle steps will always produce the same set of molecules.

Time spent:0 sec
Subject:
Biochemistry

QID:403418
System:
Metabolic Reactions

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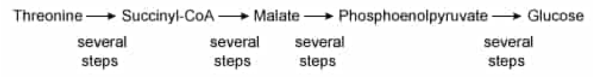

Scheme I

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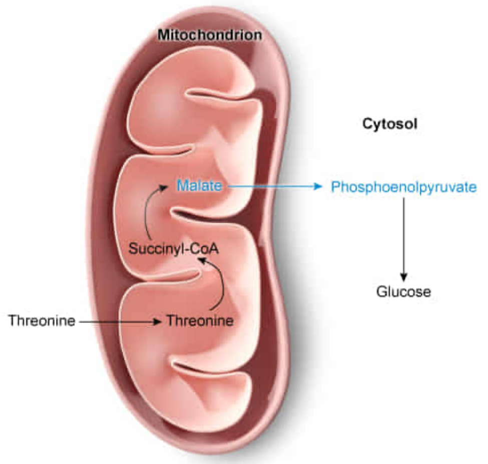
During which set of steps in Scheme I do the carbon-14 atoms exit the mitochondria and enter the cytosol?

- A. During conversion of threonine to succinyl-CoA [3%]
- B. During conversion of succinyl-CoA to malate [6%]
- ✔ C. During conversion of malate to phosphoenolpyruvate [69%]
- D. During conversion of phosphoenolpyruvate to glucose [20%]

Correct
69% answered correctly

Explanation:

Localization of metabolic processes



Metabolites exit the mitochondria during conversion of malate to phosphoenolpyruvate

Metabolic pathways are often confined to specific compartments within eukaryotic cells, allowing the cell to regulate distinct processes separately. As metabolites are converted from mitochondrial pathways to cytosolic pathways (or vice versa), they must be transported to the appropriate compartment.

The citric acid cycle occurs in the **mitochondria**, whereas gluconeogenesis occurs **primarily in the cytosol**. Scheme I shows that conversion of threonine to glucose involves several citric acid cycle intermediates, including succinyl-CoA and malate. Malate is converted to phosphoenolpyruvate through a series of steps, and then phosphoenolpyruvate undergoes gluconeogenesis, which occurs in the cytosol. Because the malate molecules undergo citric acid cycle reactions in the mitochondria and the phosphoenolpyruvate undergoes gluconeogenesis in the cytosol, the carbon-14 atoms within those molecules must **exit the mitochondria** and **enter the cytosol during this process**.

(Choice A) Succinyl-CoA is part of the citric acid cycle. As such, when threonine is converted to succinyl-CoA, its carbon-14 atoms are *inside* the mitochondria; they do not exit to the cytosol.

(Choice B) The steps that convert succinyl-CoA to malate are all part of the citric acid cycle, so the carbon-14 atoms remain in the mitochondria throughout this process.

(Choice D) The steps that convert phosphoenolpyruvate to glucose all occur in the cytosol, so the carbon-14 atoms must remain in the cytosol throughout this process.

Educational objective:
In eukaryotes, different metabolic processes occur in different cellular compartments. The citric acid cycle occurs in the mitochondria, and gluconeogenesis occurs in the cytosol.

Time spent: 0 sec
Subject: Biochemistry

QID: 403419
System: Metabolic Reactions

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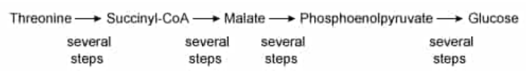

Scheme I

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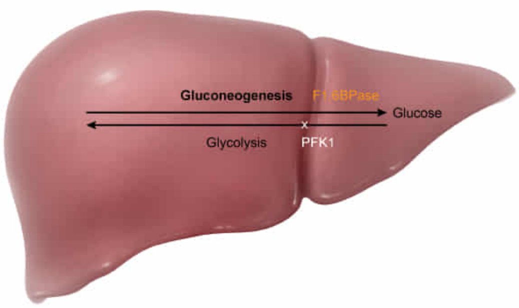
Which of the following statements correctly describes the activities of the metabolic enzymes found in the liver cells of the rats that were tested?

- A. Fructose-1,6-bisphosphatase is more active in the livers of rats that had food withheld for 3 days than for 1 day. [73%]
- B. Fructose-1,6-bisphosphatase is more active in the livers of rats that had food withheld for 1 day than for 3 days. [8%]
- C. Phosphofructokinase-1 is more active in the livers of rats that had food withheld for 3 days than for 1 day. [15%]
- D. Phosphofructokinase-1 is less active in the livers of rats that had food withheld for 1 day than for 3 days. [2%]

Correct
73% answered correctly

Explanation:

Regulatory enzymes of glycolysis and gluconeogenesis



Rats that had food withheld for 3 days conducted more gluconeogenisis.
Therefore, F1,6BPase was likely more active in these rats.

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Metabolic pathways are often regulated by activation or inhibition of specific enzymes in the pathway. This activation may happen allosterically, through hormonal control, or both. Commonly, when an enzyme in one pathway is activated, an enzyme in the opposing pathway is simultaneously inhibited. For example, glycolysis is largely regulated through the enzyme

phosphofructokinase-1, and gluconeogenesis is regulated by the opposing enzyme, **fructose-1,6-bisphosphatase**.

The information in the passage shows that rats that had food withheld for 3 days converted threonine to glucose at a faster rate than rats that had food withheld for 1 day. Therefore, rats that had food withheld for 3 days were conducting gluconeogenesis, which occurs primarily in the liver, at a faster rate. To accomplish this, those rats must have upregulated the liver enzyme that controls gluconeogenesis—fructose-1,6-bisphosphatase. Accordingly, it is likely that **fructose-1,6-bisphosphatase** was **more active** in the livers of rats that had **food withheld for 3 days** than in rats that had food withheld for 1 day.

(Choice B) If fructose-1,6-bisphosphatase were more active in the livers of rats that had food withheld for 1 day, then those rats would be expected to show a higher rate of glucose production than rats that had food withheld for 3 days. According to the passage, this is not the case.

(Choices C and D) If phosphofructokinase were more active in the livers of rats that had food withheld for 3 days, that would mean glycolysis is more active in the livers of those rats. This would result in increased glucose *consumption* in the liver relative to rats that had food withheld for 1 day, not increased glucose production. The same is true if rats that had food withheld for 1 day had *less* phosphofructokinase-1 activity than rats that had food withheld for 3 days.

Educational objective:
Glycolysis and gluconeogenesis are largely controlled by the activities of phosphofructokinase-1 and fructose-1,6-bisphosphatase, respectively.

Time spent:0 sec
Subject:
Biochemistry

QID:403420
System:
Metabolic Reactions

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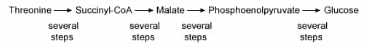

Scheme I

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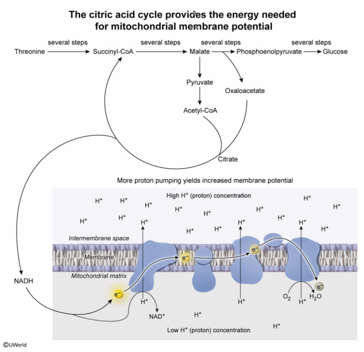
If the mitochondrial membrane potential decreased, which of the following changes to the metabolic pathway in Scheme I would best help restore the membrane potential?

- A. Malate could be converted to oxaloacetate and then to citrate. [51%]
- B. Succinyl-CoA could be converted directly to α-ketoglutarate and then to glutamate. [20%]
- C. Phosphoenolpyruvate could be converted to pyruvate and then to lactate. [20%]
- D. Glucose could be converted to 6-phosphogluconate and then to ribose-5-phosphate. [7%]

Correct
51% answered correctly

Explanation:



The mitochondrial membrane potential is an electrical potential that arises from a difference in proton concentrations between the mitochondrial matrix and the intermembrane space. This difference is achieved as the complexes of the **electron transport chain** (ETC) **pump protons** from the matrix to the intermembrane space. The energy needed to pump protons is provided by the electrons that enter the ETC, which come from the reduced cofactors NADH and $FADH_2$.

To increase the mitochondrial membrane potential in a cell, more reduced cofactors must enter the ETC. The citric acid cycle produces 3 NADH and 1 $FADH_2$ molecules per round. Therefore, **increasing the activity of the cycle** could provide the needed **cofactors** to **increase the mitochondrial membrane potential**.

Scheme I shows malate being converted to phosphoenolpyruvate (PEP), which then enters gluconeogenesis. To become PEP, malate must first become oxaloacetate. If the oxaloacetate then reacted with acetyl-CoA (most likely provided by fat stores) to become citrate (instead of becoming PEP), it would then re-enter the citric acid cycle, which would produce the needed cofactors for the ETC. Therefore, this change to the pathway in Scheme I would best help restore the mitochondrial membrane potential.

(Choice B) The citric acid cycle reaction that converts α-ketoglutarate to succinyl-CoA is irreversible, so succinyl-CoA cannot be directly converted to α-ketoglutarate. Additionally, if this reaction *did* happen, it would *consume* NADH, which would contribute to a *decreased* membrane potential. Conversion of α-ketoglutarate to glutamate would have no impact on the membrane potential.

(Choice C) Conversion of phosphoenolpyruvate to pyruvate would help increase the membrane potential *if* the pyruvate were subsequently converted to acetyl-CoA. However, converting pyruvate to lactate *consumes* NADH and would *not* help increase the membrane potential.

(Choice D) Conversion of glucose to 6-phosphogluconate and then to ribose-5-phosphate is part of the pentose phosphate pathway and would not readily affect mitochondrial membrane potential.

Educational objective:
The electric potential across the mitochondrial membrane arises as protons are pumped from the mitochondrial matrix into the intermembrane space by the electron transport chain. The citric acid cycle provides the reduced electron carriers needed for the electron transport chain and therefore helps control the membrane potential.

Time spent: 0 sec
Subject: Biochemistry

QID: 403421
System: Metabolic Reactions